BOUNDARY

Project abstract

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Boundary is an independent educational application for cricket franchise analysis and scouting. It addresses a common research problem: a decision must be supported by evidence with clear player identity, date range, competition, phase and assumptions. The project combines a Streamlit interface, structured match analytics, constrained selection and bowling scheduling, an optional DeepSeek fact-selection step, and reproducible decision exports.

The supplied frozen data contains 6,392 eligible men's matches, 141,377 player-match records and 6,618 archive player identities across IPL, BBL, The Hundred, T20I and SMAT. Deterministic Python tools calculate descriptive batting and bowling scores. A mixed-integer optimizer chooses eleven players and assigns all twenty overs under stated workload, role, availability and budget constraints. DeepSeek can select only existing fact IDs and approved next actions; it cannot create statistics or change calculation controls.

Local evaluation includes 68 automated tests, 57 browser checks and independent reconciliation of 3,393,048 numeric fields against the supplied delivery archives, with zero ingestion mismatches. A separate sample of 194 official-source numeric checkpoints produced 188 matches, five measured coverage gaps and one unresolved conflicting figure. A chronological research forecast failed its stronger-baseline gate, so the application displays the historical baseline. The result is an inspectable capstone research tool. Complete official coverage, full current-roster verification, demonstrated tactical benefit and production team access remain future work.

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Source: https://github.com/br-abishek-07/boundary-cricket-capstone
Keywords: cricket analytics, evidence provenance, constrained AI, DeepSeek, mixed-integer optimization, Streamlit.